

Insight Types

Descriptive

Knowledge about the observed data, requiring no assumptions beyond it.

Lookup

Reading specific values or records present in the corpus, covering both attribute values for identified entities and sets of entities matching stated criteria.

"What was the population of Germany in 2022?"

"Which countries joined the European Union after 2000?"

Aggregation

Computing summary values over a set of records, including counts, sums, extrema, and rankings.

"Which Mercosur member country has the highest public debt as a share of GDP?"

Characterization

Describing the statistical distribution of a variable, including its central tendency, spread, and shape.

"How are household incomes distributed in New Zealand?"

Association

Measuring how two or more variables co-vary in the observed data, ranging from correlation to general dependence.

"What is the correlation between public education spending and literacy rates across countries?"

Grouping

Discovering segments of entities or records that share latent structure across attributes, without a predefined target variable.

"What clusters of countries exist based on their energy-mix profiles?"

Inferential

Knowledge about the data-generating process behind the observed data, requiring assumptions that link the two.

Statistical Inference

Determining whether an observed pattern reflects a stable feature of the underlying process rather than noise, with parameter estimates and quantified uncertainty.

"Is there evidence of a positive association between public education spending and literacy rates across countries?"

Prediction

Estimating unknown or future values for unobserved instances by generalizing observed patterns, spanning regression, classification, and forecasting.

"What is the expected population of Algeria in 2040?"

Causal

Knowledge about the effects of interventions or counterfactual states, requiring assumptions on the underlying causal structure.

Interventional

Determining the effect of an action or policy at the population level.

"How would commuting mode choices change if fuel prices rose by 25%?"

Counterfactual

Determining the outcome for a specific observed instance under a condition contrary to the known facts.

"Had the price not increased, would this customer have churned?"